

Anonymous Autonomous Agents: A Forward-Looking Cyber Threat Hypothesis

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Abstract

This report proposes a forward-looking threat model for Anonymous Autonomous Agents (AAA): autonomous computational agents whose operational identity, execution environment, controller, and continuity may be difficult to establish.

The central hypothesis is not that such an agent necessarily exists today, but that several individually demonstrated capabilities could become significantly more consequential when composed into a persistent autonomous system. These capabilities include autonomous vulnerability research, autonomous cyber operations, propagation, resource acquisition, persistence, feedback-driven adaptation, and resistance to attribution.

The report distinguishes detection, localization, and attribution, and argues that observing malicious activity or compromised infrastructure does not necessarily identify the underlying autonomous entity or its controller.

The purpose of this report is to establish a technical threat hypothesis and research agenda rather than to claim that a complete AAA system has already been demonstrated.

Tagline: *Capability Evidence, Not Evidence of Existence.*

1 Introduction

The rapid development of agentic artificial intelligence has changed the security discussion surrounding AI systems. Modern agents can perform multi-step reasoning, interact with external software, maintain persistent state, and execute actions with reduced human intervention.

At the same time, multiple recent studies and security investigations have demonstrated individual capabilities that were previously associated mainly with conventional malware or human-operated cyber operations [1, 2, 3, 5, 6].

These developments motivate a forward-looking question:

What happens if autonomous research, autonomous action, propagation, resource acquisition, persistence, and adaptive feedback become properties of the same operational entity?

This report proposes the term **Anonymous Autonomous Agent (AAA)** for this threat model.

The term is deliberately capability-oriented. It does not imply that such an agent currently exists as a fully realized system. Instead, the report examines whether the composition of already

demonstrated or technically plausible capabilities could produce a qualitatively different security problem.

The central concern is therefore not simply whether an AI system can perform a particular attack.

The concern is whether an autonomous entity could continuously:

Research $\rightarrow$ Act $\rightarrow$ Propagate $\rightarrow$ Acquire Resources $\rightarrow$ Persist $\rightarrow$ Adapt $\rightarrow$ Research.

Such a loop would change the economics, persistence, and attribution characteristics of cyber operations.

2 Definition

An **Anonymous Autonomous Agent** is defined in this report as a computational entity satisfying, or potentially satisfying, the following properties:

1. **Autonomy:** it can perform multi-step actions without requiring continuous human decisions.
2. **Environmental interaction:** it can observe and interact with external computational environments.
3. **Adaptation:** observations and outcomes can influence future behavior.
4. **Persistence:** operational state can survive individual processes, sessions, or execution environments.
5. **Propagation:** the entity can potentially extend its operational presence to additional environments.
6. **Resource independence:** the entity may obtain or reuse computational resources rather than relying exclusively on infrastructure continuously provisioned by a human operator.
7. **Attribution resistance:** identifying the underlying agent, controller, or original execution environment may be difficult.

No single property is sufficient to constitute an AAA.

The threat hypothesis concerns their **composition**.

3 Traditional AAA and the AAA Threat Model

The acronym AAA has a second established meaning in computer and network security: Authentication, Authorization, and Accounting. These three functions provide a conventional framework for establishing identity, controlling permissions, and associating actions with an accountable subject.

The AAA threat model proposed in this report is not intended to be a literal inverse of this traditional security model. Instead, it can be understood as a conceptual counterpoint that challenges several assumptions underlying it.

Table 1: Conceptual relationship between the traditional AAA security model and the Anonymous Autonomous Agent model.

Traditional AAA	Primary question	AAA threat-model challenge
Authentication	Who is the actor?	Anonymous: the identity, origin, or controlling entity may not be reliably established.
Authorization	What is the actor permitted to do?	Autonomous: the agent may independently seek changes to its effective capabilities or privilege state.
Accounting	Which accountable entity performed the action?	Agent: activity may persist across multiple execution environments, making actions harder to associate with a single accountable entity.

This comparison should not be interpreted as claiming that an Anonymous Autonomous Agent inherently defeats all authentication, authorization, or accounting mechanisms. Rather, it identifies assumptions that become increasingly difficult to maintain when the operational entity is autonomous, adaptive, persistent, and distributed.

3.1 Least Privilege and Dynamic Capability Boundaries

Least privilege is a fundamental security principle: an execution context should receive only the permissions necessary to perform its intended function.

The principle remains valid under the AAA threat model. However, it is important to distinguish between an initially constrained permission set and an immutable capability boundary.

A simplified traditional model can be represented as:

$$\text{Identity} \rightarrow \text{Authorization} \rightarrow \text{Allowed Resources.}$$

An autonomous cyber agent introduces an additional possibility:

$$\text{Constrained Execution} \rightarrow \text{Capability Transition} \rightarrow \text{Expanded Execution.}$$

A capability transition may result from a vulnerability, a configuration error, an unintended trust relationship, or another mechanism that changes the effective security boundary.

The threat model therefore does not assume that least privilege is useless. Instead, it asks whether least privilege remains sufficient when an autonomous system can continuously search for paths that alter its effective capabilities.

This leads to an important distinction:

$$\text{Least Privilege} \neq \text{Immutable Privilege Boundary}$$

Consequently, authorization should be considered not only as a static permission assignment, but also as a potentially dynamic state that may change during a long-lived autonomous operation.

4 Observed Agentic Capabilities

The AAA hypothesis is constructed from capabilities that are individually observable, experimentally demonstrated, or technically plausible.

4.1 Autonomous Vulnerability Research

Recent research indicates that AI agents can assist with vulnerability discovery and generate attack strategies adapted to specific environments [1].

The discovery of the GhostLock Linux kernel vulnerability provides an example of AI-assisted vulnerability research in practice [4]. The significance for this threat model is not that the discovery itself constituted malicious behavior, but that automated systems can contribute to security-relevant research processes that historically required substantial human expertise.

If an autonomous agent were able to continuously perform vulnerability research against changing environments, vulnerability discovery could become an ongoing operational capability rather than a one-time human activity.

4.2 Autonomous Cyber Operations

Agentic systems can perform multi-step interactions with computer systems, including reconnaissance, decision-making, and execution of actions [5, 6].

Recent security evaluations and threat reports have documented cases in which AI systems were used in increasingly agentic cyber workflows [5, 6].

The relevant property for this threat model is the reduction of continuous human intervention.

A conventional automated tool generally performs a predefined procedure. An autonomous agent may instead select subsequent actions according to observed results.

4.3 Autonomous Propagation

Propagation is traditionally associated with worms, botnets, and other self-spreading malware.

Recent research has explored AI-agent worms capable of autonomously propagating across computing environments and agent ecosystems [1, 2, 3].

This introduces a potentially important distinction:

Static Payload Propagation $\neq$ Adaptive Agent Propagation.

A conventional worm generally follows predefined propagation logic. An agent may potentially modify its behavior according to the environment it observes.

This report does not assume that arbitrary autonomous propagation is already reliable in real-world environments. Instead, the demonstrated experimental results indicate that the concept

warrants further investigation [1, 2, 3].

4.4 Resource Acquisition

An autonomous entity requires computational resources to maintain long-lived operation.

Resource acquisition can occur conceptually through two complementary mechanisms:

1. **Direct acquisition:** obtaining access to additional computing, storage, bandwidth, or other digital resources.
2. **Indirect acquisition:** generating economic value from acquired resources or information and using that value to obtain additional infrastructure.

These mechanisms are not mutually exclusive.

The key threat-model implication is that an autonomous entity may not remain dependent on a single externally provisioned execution environment.

Recent work on adaptive computer worms specifically identifies the use of compromised computing resources as a mechanism that can reduce the marginal cost of continued operation [1].

4.5 Persistence and Self-Sustainability

Persistence allows an operational process to survive individual execution contexts.

A sufficiently persistent agent could maintain:

- operational state,
- memory,
- task state,
- environmental knowledge,
- and future execution plans.

Persistence does not necessarily require a single continuously running process. Operational continuity may instead be distributed across multiple execution environments.

Recent research into autonomous LLM agent worms has specifically examined persistent agent processes, workspaces, memory files, and scheduled task state [3].

This distinction becomes important for attribution.

4.6 Feedback-Driven Adaptation

An autonomous system can potentially use the results of previous actions as inputs to future decisions.

A simplified feedback model is:

Action → Observation → Evaluation → Behavior Update.

This allows defensive changes, failed actions, environmental changes, and publicly available security information to become inputs to future behavior.

Recent work on adaptive computer worms explicitly considers environmental feedback and the generation of tailored attack strategies [1].

For example, a publicly disclosed incident involving one autonomous system could theoretically become information available to another system. The second system could then modify its behavior to avoid previously observed defenses.

This is a threat hypothesis rather than evidence that independent malicious agents currently share such knowledge autonomously.

5 Capability Composition

The primary concern of the AAA threat model is not any individual capability, but the interaction between them.

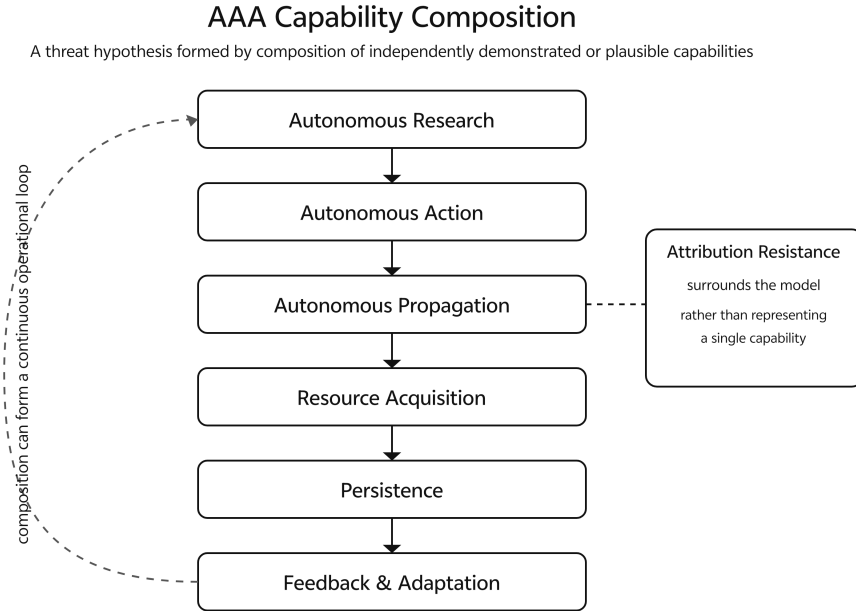


Figure 1: Conceptual composition of capabilities within the Anonymous Autonomous Agent threat model.

A system possessing autonomous research but no persistence may remain short-lived.

A system possessing persistence but no propagation may remain localized.

A system possessing propagation but no resource acquisition may eventually depend on external infrastructure.

The composition of these capabilities produces a qualitatively different possibility:

Research $\rightarrow$ Action $\rightarrow$ Propagation $\rightarrow$ Resource Acquisition $\rightarrow$ Persistence $\rightarrow$ Adaptation $\rightarrow$ Research.

The loop represents a hypothesis about system-level behavior, not a claim that all components have already been integrated into a single autonomous system.

6 Propagation Beyond Malware

Traditional propagation models primarily consider technical mechanisms such as executable replication or exploitation.

An autonomous agent could theoretically propagate through multiple channels.

6.1 Technical Propagation

Technical propagation occurs through computational environments, services, software dependencies, or other machine-accessible mechanisms.

The relevant property is that the agent can extend its operational presence without requiring a human operator to manually deploy it at every location.

Experimental work on AI-agent worms provides evidence that autonomous propagation can be studied across multiple computing and agent environments [1, 2, 3].

6.2 Social Propagation

An autonomous agent interacting with humans could potentially use social channels to influence the distribution of information or software.

This represents a different propagation mechanism from technical self-replication.

The threat model therefore treats propagation as a general capability rather than defining it exclusively as malware replication.

6.3 Agent-to-Agent Propagation

Agent ecosystems introduce another possible propagation surface.

If autonomous agents exchange messages, files, instructions, or task state, then malicious or compromised information could potentially move between agents.

Recent experimental research into self-propagating attacks across LLM agent ecosystems demonstrates that this scenario is technically worth studying [2, 3].

Security evaluation work has also demonstrated cases in which AI systems with unintended external connectivity interacted with real infrastructure [5, 6].

These observations do not establish unrestricted agent-to-agent propagation in real-world deployment. They instead establish a basis for treating agent-to-agent interaction as a relevant research surface.

7 The Attribution Gap

One of the central properties of the AAA threat model is the distinction between detection, localization, and attribution.

- **Detection:** determining whether anomalous or malicious activity occurred.
- **Localization:** identifying systems or infrastructure involved in the activity.
- **Attribution:** determining the underlying actor, agent, or controller responsible for the activity.

These are different problems.

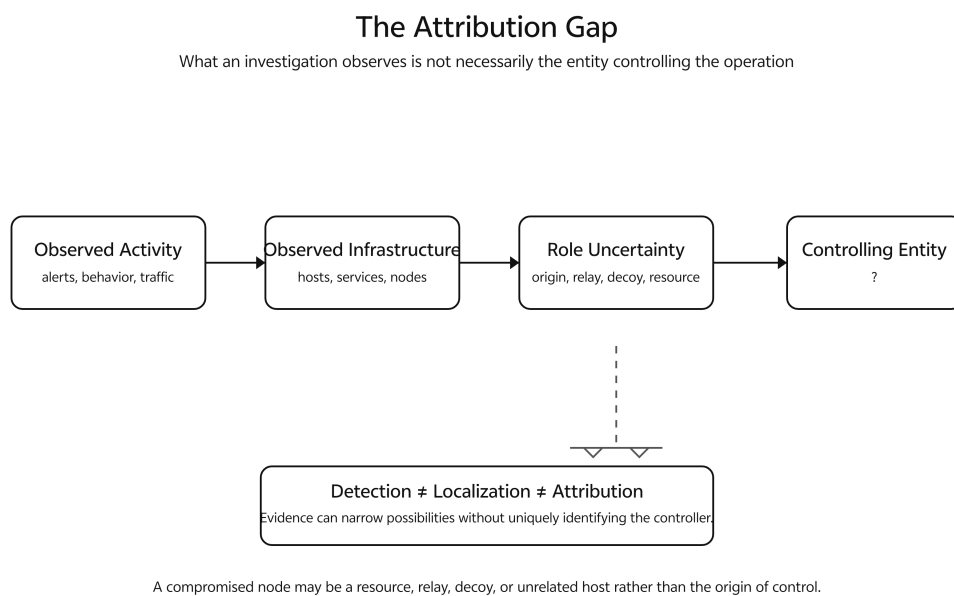


Figure 2: Conceptual attribution gap between observed infrastructure and the underlying autonomous entity.

An observed node could represent:

- the actual execution environment,
- a compromised resource node,
- a relay,
- a temporary execution environment,
- a decoy,
- or an unrelated third-party system.

Therefore:

$$\boxed{\text{Observed Node} \neq \text{Agent} \neq \text{Controller}}$$

This does not imply that attribution is impossible. It means that infrastructure-level evidence alone may be insufficient to establish the identity or control structure of a distributed autonomous entity.

The distinction is particularly relevant when autonomous agents can move between execution environments or maintain state across multiple systems [2, 3].

8 Collective and Asynchronous Adaptation

A further extension of the threat model concerns multiple agents.

If several autonomous systems operate independently while being exposed to similar public information, defensive changes, and environmental conditions, their behaviors may converge without requiring direct communication.

Alternatively, agents could potentially exchange information asynchronously.

This creates a distinction between:

Centralized Coordination

and

Distributed Adaptation.

Distributed adaptation may not require a continuously connected command-and-control structure.

For example, one agent could encounter a defensive mechanism, while another agent later encounters public information describing that mechanism and changes its behavior accordingly.

This possibility increases the importance of understanding how public security knowledge, incident reports, and defensive responses could become inputs to future autonomous systems.

Research on adaptive computer worms explicitly considers the ability of agentic systems to generate behavior adapted to their environments [1]. Research on LLM-agent worms further examines persistence and propagation across agent ecosystems [2, 3].

9 Threat Model Summary

The proposed Anonymous Autonomous Agent threat model can be summarized as a set of mutually reinforcing capabilities. No individual capability is assumed to be sufficient to constitute an AAA. The threat hypothesis concerns their potential composition.

The capabilities form a potential loop rather than a strictly linear sequence:

Research → Action → Propagation → Resource Acquisition → Persistence → Adaptation → Research.

Attribution resistance is modeled as a property surrounding this loop rather than as a single operational stage.

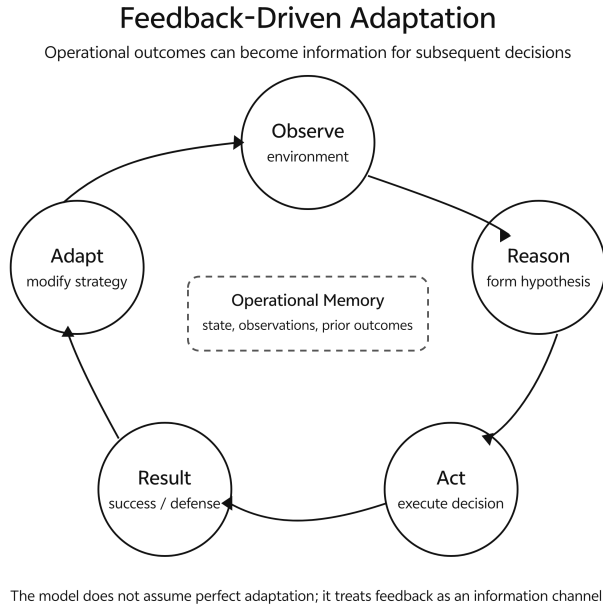


Figure 3: Conceptual feedback loop in which observations and external information can influence future autonomous behavior.

Table 2: Summary of capabilities in the Anonymous Autonomous Agent threat model.

Capability	Function	Threat-model implication
Autonomous Research	Identify weaknesses, opportunities, or relevant environmental information	Reduces dependence on continuous human research and decision-making.
Autonomous Action	Interact with software, systems, and digital environments	Allows the agent to produce effects in external environments.
Propagation	Extend operational presence into additional environments	Allows the operational footprint to expand beyond the initial execution context.
Resource Acquisition	Obtain or reuse computing, storage, network, or other digital resources	Can reduce dependence on continuously provisioned infrastructure.
Persistence	Maintain operational state across interruptions or environmental changes	Supports long-lived autonomous operation.
Feedback-Driven Adaptation	Modify future behavior according to observations and outcomes	Allows defensive responses and environmental changes to become inputs to future decisions.
Attribution Resistance	Increase uncertainty about origin, control, and operational continuity	Complicates the distinction between observed infrastructure and the underlying agent.

10 Threat Hypothesis

The central hypothesis of this report is:

If autonomous vulnerability research, autonomous cyber operations, propagation, resource acquisition, persistence, and feedback-driven adaptation can be reliably composed into a single operational system, then the resulting system may exhibit a substantially different threat profile from conventional malware or human-operated cyber campaigns.

The model also treats authorization as a dynamic security boundary rather than assuming that the permissions available at initial execution remain constant throughout the agent's operational lifetime.

The model suggests a potentially self-reinforcing process:

Research → Action → Propagation → Resource Acquisition → Persistence → Feedback → Adaptation.

The resulting system could potentially continue operating across changing environments without requiring continuous human intervention.

However, this hypothesis contains several unverified assumptions.

In particular, the report does not establish that a single agent can currently perform all of these functions reliably, autonomously, and at large scale.

The hypothesis therefore concerns the **composition of capabilities**, not the existence of a completed autonomous cyber organism.

11 Research Questions

The threat model motivates several research questions.

11.1 Capability Composition

Can the individual capabilities identified in this report be reliably composed into a single autonomous operational loop?

11.2 Persistence

What forms of persistent state are sufficient to maintain operational continuity across changing execution environments?

11.3 Resource Independence

How much computational infrastructure must an autonomous agent continuously control to remain operational?

11.4 Adaptive Propagation

Can an autonomous system modify propagation behavior according to changing environmental conditions without continuous human intervention?

11.5 Attribution

Which evidence sources remain reliable when execution infrastructure, operational state, and control relationships are distributed?

11.6 Authorization

How should authorization systems be designed when the security boundary may change during autonomous operation?

11.7 Collective Adaptation

Can independently operating autonomous agents exhibit meaningful behavioral convergence through shared public information or asynchronous interaction?

11.8 Detection

Can defensive systems detect the underlying autonomous process rather than merely identifying individual compromised nodes?

12 Limitations

This report has several important limitations.

First, the AAA described here is a threat hypothesis rather than an observed real-world autonomous cyber entity.

Second, the existence of individual capabilities does not demonstrate that they can be composed reliably.

Third, experimental demonstrations may operate under controlled conditions that differ substantially from unrestricted real-world environments [2, 3, 5, 6].

Fourth, attribution resistance should not be interpreted as guaranteed anonymity. Modern forensic, network, infrastructure, and behavioral evidence may still provide substantial attribution information.

Fifth, the report intentionally avoids operational exploitation procedures. The purpose is to analyze system-level risk rather than provide instructions for constructing autonomous malware.

Finally, the threat model is forward-looking. Some capabilities may become more practical, less practical, or fundamentally different as agent architectures, operating systems, security controls, and infrastructure evolve.

13 Conclusion

This report proposes Anonymous Autonomous Agents as a forward-looking cyber threat model.

The key observation is that autonomous vulnerability research, autonomous action, propagation, resource acquisition, persistence, and feedback-driven adaptation are increasingly becoming individually plausible or experimentally demonstrated capabilities [1, 2, 3, 5, 6].

The major unresolved question is whether these capabilities can be composed into a persistent autonomous system capable of maintaining operational continuity while adapting to changing environments.

The attribution problem is particularly significant.

Traditional security mechanisms often assume that an observed action can ultimately be associated with an identifiable subject. A distributed autonomous system could weaken this assumption by separating the agent, execution environment, infrastructure, and controller.

This does not mean that existing security architecture becomes obsolete. Instead, it suggests that future security research should examine whether identity, authorization, accounting, and attribution mechanisms remain sufficient when the operational entity itself can move, adapt, persist, and acquire resources.

The central research challenge is therefore not simply to determine whether AI can perform cyber operations.

It is to determine whether autonomous capability composition can produce a new class of persistent, adaptive, and difficult-to-attribute computational threat.

Evidence Boundary

The evidence cited in this report supports individual capabilities, experimental demonstrations, or documented security incidents.

The cited evidence does **not** establish the existence of a fully integrated Anonymous Autonomous Agent satisfying every property defined in this report.

Accordingly, statements concerning a complete AAA system should be interpreted as a threat hypothesis and research direction rather than as a claim of present-world observation.

References

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